



# S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part – 2 (Paper-3)

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## Functions in C++

- A function is a group of statements that together perform a task. Every program in C++ has **main()** function by default. C++ standard library provides lot of built in functions that you can call or use in your program.

OR

- **A function is a block of code that performs a specific task and can be reuse anywhere in the program.**
- All function Parentheses ( ) are present after the name  
Example: add(), hira(), main(),.....
- Syntax :

```
return_type function_name(parameters)
{
    // function body
}
```

- Example :

```
int add(int a, int b)
{
    return a + b;
}

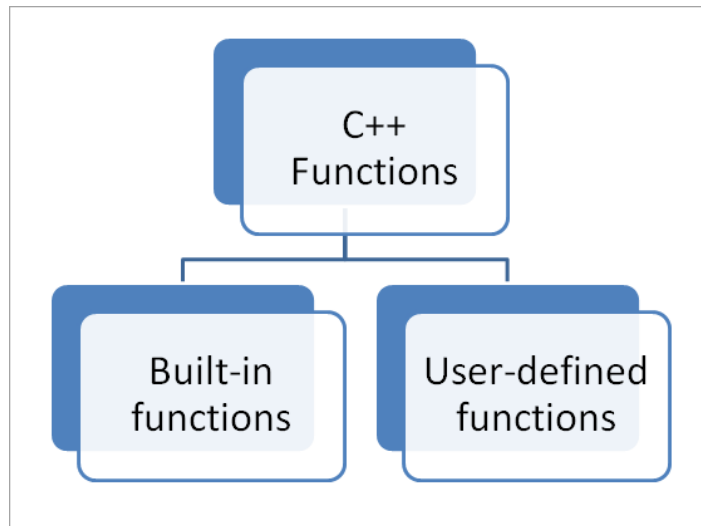
int main()
{
    cout << add(10, 20);
}
```

Output: 30

**There are two types of function:**

# 1. Standard Library Functions / Predefined Function / Built in Functions

## 2. User-defined Function / Custom Function



## 1. Predefine Function

- Predefine functions are the functions that are already written and stored in c++ standard libraries. Programmers can directly use them in their program without writing their definition.
- You only need to include the correct header file to use them.

OR

- A predefined function in C++ is a function that is already define in the C++ standard library and can be used directly by including the appropriate. header file.

- **Syntax**

```
#include <header_file>  
function_name(arguments);
```

- **Example**

```
#include <cmath>  
int x =sqrt(25);
```

## 2. User define Function

- A User defined function in C++ is a function that is define by the programmer to perform a specific task and can be called whenever needed in the program.

- **Syntax**

```
void hira()  
{  
    // code to be executed
```

}

Here, **hira()** is the name of the function.

**void** means that the function does not have a return value.

inside the function (the body), **add code** that defines what the function should do.

## ❖ Header Files

- Contains a collection of function prototypes, constant and preprocessor definitions
- Named with extension .h

A header file may be included in one of the two ways.

# include <iostream.h>                      (use angle bracket <> )or  
# include "iostream.h".                      (use double quotes " " )

- ✓ The header file in angle brackets means that file reside in standard include directory (predefined files).

e.g.- #include <iostream>  
      #include <cmath>  
      #include <cstring>,.....

- ✓ The header file in double quotes means that file resides in current directory (User defined files).

e.g. -        #include "hira.h"  
              #include "Bihar.h",.....

## ❖ Preprocessor

- Preprocessor commands begin with a # and do process of instruction available in #.

e.g.        #include – includes a named file  
              #define – defines a (text replacement) macro  
              #ifdef/#else/#endif – conditional compilation

- ❖ **# define** = Often used to define constants.

e.g. #include <iostream>

```
#define PI 3.14      //define constant & PI (identifier) is a macro  
# const float PI = 3.14;      // define constant
```

## #define vs constants

| #define                                                                                                                            | Constants                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| It has no data type<br>No memory is allocated<br>It is replaced by the preprocessor before compilation.<br>Debugging is difficult. | It has a proper data type (float)<br>Memory is allocated<br>It is handled by the compiler.<br>It is safer and more modern. |

## ❖ List of Predefine Function header files in c++ and its use/work

### 1. <iostream>

Used for input and output operations in C++.

| Name             | Use                                                          | Type           |
|------------------|--------------------------------------------------------------|----------------|
| <b>cout</b>      | Display normal output on the screen(Standard output)         | ostream object |
| <b>cin</b>       | Takes input from the keyboard without space (standard input) | istream object |
| <b>getline()</b> | Reads an entire line, including spaces                       | Function       |
| <b>endl</b>      | Inserts a new line and flushes the output buffer             | Manipulator    |
| <b>ws</b>        | Consumes leading whitespace from a input stream              | manipulator    |
| <b>flush</b>     | Flushes the output buffer                                    | manipulator    |

- **Manipulator** : A manipulator is a special facility used to control / modify the formatting or behavior of an input/output stream. Manipulator is normally used with an I/O stream, such as cout or cin.
- **Buffer memory** : Buffer memory is a temporary storage area that holds data until it is ready to be processed or transferred.

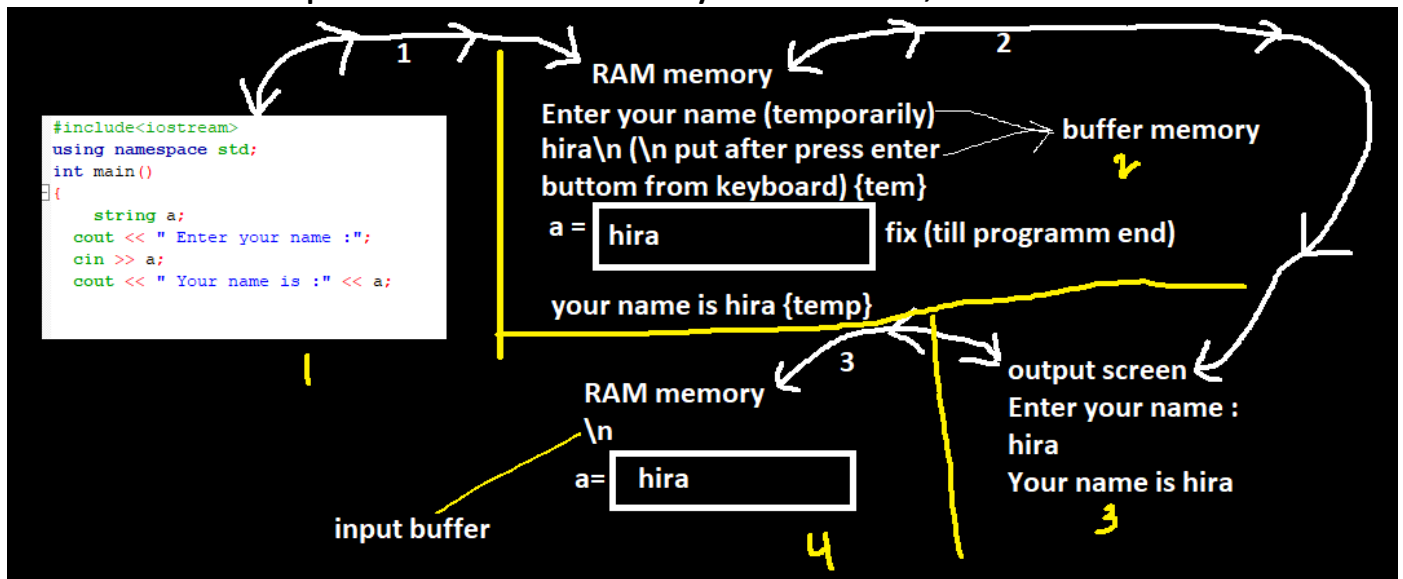
Two types of buffer memory

(i) **Input buffer** : Temporarily stores data received from an input device, such as the keyboard before the program reads it.

Example : `cin << a;`

(ii) **Output buffer** : Temporarily stores output data before it is sent to the screen or another output device.

Example : `cout << "enter your name";`



Note : मान लीजिए आप लिखते हैं:

hira kumar[Enter]

Keyboard से input stream (stream means data flow process/channel ) में लगभग यह sequence आता है:

h i r a k u m a r \n

यहाँ:

- hira → characters
- space → space character
- kumar → characters
- Enter → \n (newline)

इसलिए जब आप: `cin >> s;` करते हैं, तो cin केवल पहला word पढ़ता है:

`s = "hira"`

लेकिन बाकी input buffer में रहता है:

`bkumar\n`

फिर अगर: `cin.ignore();` लगाते हैं, तो वह सिर्फ पहला character (space) हटाता है: `kumar\n` रह जाता है |

और:

`getline(cin, city);` → `kumar` पढ़ लेता है।

अगर आप चाहते हैं कि input buffer में मौजूद पूरा बाकी all input clear हो जाए, तो `cin.ignore()` को arguments के साथ use कर सकते हैं:

`cin.ignore(1000, '\n');`

इसका मतलब:

अधिकतम 1000 characters ignore करो, या जब \n मिल जाए तो रुक जाओ।

या Better/general form

`cin.ignore(numeric_limits<streamsize>::max(), '\n');`

इसके लिए: `#include <limits>` header file का use करना चाहिए।

➤ **Flush** : Send the buffered data (buffer memory data) to its destination immediately (output screen,...)

```
hello.cpp X C:\Users\hira\Desktop\hello.exe
#include<iostream>
using namespace std;
int main()
{
    string s,city;
    cout << " Welcome C++ programming language";
    cout << endl;
    cout << "what is your name :";
    cin >> s;
    cout <<" your name is : " << s << flush;
    cout <<flush;
    cout << endl << "what is your city name : ";
    cin.ignore( 1000, '\n'); // ignore \n
    getline (cin, city);
    cout << "your city name is : " << city;
}
```

```
Welcome C++ programming language
what is your name :hira kumar
your name is : hira
what is your city name : bihar sharif
your city name is : bihar sharif
Process returned 0 (0x0) execution time : 5.988 s
Press any key to continue.
```

## 2. <cmath> or <math.h>

<cmath> use modern c++ preferred.<math.h> old c++ also support.  
<cmath> is a standard c++ header file that provides mathematical functions for performing calcuations.

### Basic mathematical functions

| Function name | Use                                | Example              |
|---------------|------------------------------------|----------------------|
| abs()         | Absolute value                     | abs(-10) = 10        |
| fabs()        | Absolute value of floating number  | fabs(-10.5)=10.5     |
| fmod()        | Floating point remainder           | fmod(10.5,3)=1.5     |
| fmax()        | Returns larger floating/int value  | fmax(10.5,20.5)=20.5 |
| fmin()        | Returns smaller floating/int value | fmin(10.5,20.5)=10.5 |

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
    cout <<"abs(-10 \t =" << abs(-10) << endl;
    cout << "fabs(-10.5)\t =" << fabs(-10.5) << endl;
    cout << "fmod (10.5,3)\t =" << fmod(10.5,3) << endl;
    cout <<"fmax(10.5,20.5)\t =" << fmax(10.5,20.5) << endl;
    cout << "fmin(10.5,20.5)\t =" << fmin(10.5,20.5) << endl;
    cout << "fmin(58,45)\t =" << fmin(58,45) << endl;
    cout <<"smaller value of 25,98,46 is : " << fmin(25,(fmin(98,46))) << endl;
}
```

```
C:\Users\hira\Desktop\hello.exe
abs(-10)          =10
fabs(-10.5)       = 10.5
fmod (10.5,3)     =1.5
fmax(10.5,20.5)   =20.5
fmin(10.5,20.5)   =10.5
fmin(58,45)       =45
smaller value of 25,98,46 is : 25
```

### Questions

1. Write a C++ program to find the absolute value of -125 using abs().
2. Write a C++ program to find the absolute value of -45.75 using fabs().
3. Write a C++ program to find the remainder when 25.5 is divided by 4.2 using fmod().
4. Write a C++ program to find the largest value among 25, 98, and 46 using fmax().

5. Write a C++ program to find the smallest value among -15.5, 20.75, and -8.25 using fmin().
6. Write a C++ program to find the difference between the largest and smallest values among -25.5, 40.75, -10.25, and 60.5.
7. Write a C++ program to find the remainder of the difference between -85.5 and 32.25 when divided by 7.5.
8. Write a C++ program to find the smallest absolute value among -45, 28, -12, and 35.
9. Write a C++ program to find the largest absolute value among -125.75, 98.5, -150.25, and 75.25.
10. Write a C++ program to find the remainder of the largest value divided by the smallest value among -18.5, 25.75, -32.25, and 12.5.

**Answer : 125,45.75,0.3,98,-15.5,86.25,5.25,12,150.25,25.75**

## Power and root

| Function name | Use                           | Example       |
|---------------|-------------------------------|---------------|
| sqrt()        | Square root                   | sqrt(25) = 5  |
| cbrt()        | Cube root                     | cbrt(27) = 3  |
| pow()         | Raises a number to a power    | pow(2,3)=8    |
| hypot()       | Calculates $\sqrt{a^2 + b^2}$ | hypot (3,4)=5 |

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
    cout << "square of 5 is \t = " << 5*5 << endl;
    cout << "square root of 25 is \t = " << sqrt(25) << endl;
    cout << "cube root of 27 is \t = " << cbrt(27) << endl;
    cout << "2 power 3 \t\t = " << pow(2,3) << endl;
    cout << " square root of (3 saure + 4 square) \t = " << hypot(3,4);
}
```

```
C:\Users\hira\Desktop\hello.exe
square of 5 is          = 25
square root of 25 is    = 5
cube root of 27 is      = 3
2 power 3               = 8
square root of (3 saure + 4 square)    = 5
```

## Rounding functions

| Function name | Use                                                                                             | Example       |
|---------------|-------------------------------------------------------------------------------------------------|---------------|
| ceil()        | Rounds upward                                                                                   | ceil(4.2) = 5 |
| floor()       | Rounds downward                                                                                 | floor(4.8) =4 |
| round()       | Rounds to nearest integer                                                                       | round(4.6) =5 |
| trunc()       | Removes fractional part                                                                         | trunc(4.8)=4  |
| rint()        | Rounds a/c to current rounding mode (nearest even when .5)<br>{otherwise same round() function} | rint(4.8) =4  |

```

#include<iostream>
#include<cmath>
using namespace std;
int main()
{
cout << " ceil(4.2) \t = " << ceil(4.2) << endl;
cout << " floor(4.8) \t = " << floor(4.8) << endl;
cout << " round(4.6) \t = " << round(4.6) << endl;
cout << " round(4.5) \t = " << round(4.5) << endl;
cout << " round(4.1) \t = " << round(4.1) << endl;
cout << "rint(4.6) \t = " << rint(4.6) << endl;
cout << "rint(5) \t = " << rint(5) << endl;
cout << "rint(4.1) \t = " << rint(4.1) << endl;
cout << "rint(4.5) \t = " << rint(4.5) << endl;
}

```

C:\Users\hira\Desktop\hello.exe

```

ceil(4.2)           = 5
floor(4.8)          = 5
round(4.6)          = 5
round(4.5)          = 5
round(4.1)          = 4
rint(4.6)           = 5
rint(5)             = 5
rint(4.1)           = 4
rint(4.5)           = 4

```

## Questions

- Write a C++ program to calculate  $2^8$  and round the result to the nearest integer.
- Write a C++ program to calculate the square root of 245.75 and round the result to the nearest integer.
- Write a C++ program to calculate  $5^{3.5}$  and round the result to the nearest integer.
- Write a C++ program to calculate the cube root of 125.75 and round the result to two different integer values using floor() and ceil().
- Write a C++ program to calculate the square root of  $3^8$  and display the result using round().
- Write a C++ program to calculate  $12^{2.5}$ , find its square root, and round the final result to the nearest integer.
- Write a C++ program to calculate the fifth root of 2500 using a power function and round the result to the nearest integer.
- Write a C++ program to calculate  $\sqrt{1250.5}$ , then display its rounded-down, rounded-up, and nearest integer values.
- Write a C++ program to calculate  $7^{4.2}$ , find its square root, and display the final result using rint().
- Write a C++ program to calculate the cube root of 9876.54, square the result, and display the final value using floor(), ceil(), and round().

**Answer : 256,16,280,(5, 6),81,49,5,(35, 36, 35),98,(98, 99, 98)**

# Trigonometric functions

| Function name | Use     | Example    |
|---------------|---------|------------|
| sin()         | Sine    | sin(0) = 0 |
| cos()         | Cosine  | cos(0)=1   |
| tan()         | Tangent | tan(0) =0  |

Note : Sine (x will be insert in radian value not degree)

$$\text{Radian} = \text{degree} \times \frac{\pi}{180} \quad (\pi = 3.141)$$

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
//cout <<"value of sin(0) = " << sin(0) << endl; (no, it is wrong)
cout << " value of sin(0) = " << sin(0*3.14/180) << endl;
cout << " value of cos(0) = " << cos(0*3.14/180) << endl;
cout << " value of tan(0) = " << tan(0*3.14/180) << endl;
cout << " value of cosec(0) = " << 1/sin(0*3.14/180) << endl;
```

C:\Users\hira\Desktop\hello.exe

```
value of sin(0) = 0
value of cos(0) = 1
value of tan(0) = 0
value of cosec(0) = inf
```

# Other functions

| Function name    | Use                       | Example                 |
|------------------|---------------------------|-------------------------|
| Exp()            | Calculates e <sup>x</sup> | Exp(1) = 2.71828        |
| Log()            | Natural logarithm         | Log(1) = 0              |
| Log10()          | Base-10 logarithm         | Log10(100) =2           |
| Isgreater()      | Checks > (output 0,1)     | Isgreater(10,5)=1       |
| Isless()         | Checks <(output 0,1)      | Isless(10,5) =0         |
| Isgreaterequal() | Checks>=                  | Isgreaterequal(10,10)=1 |

```
#include<iostream>
#include<cmath>
using namespace std;
int main()
{
cout << " exp(1) = " << exp(1) << endl;
cout << "log(1) = " << log(1) << endl;
cout << "log10(100) = " << log10(100) << endl;
cout << "isgreater(10,5) = " << isgreater(10,5) << endl;
cout << "isless(10,5) = " << isless(10,5) << endl;
cout << "isgreaterequal = " << isgreaterequal(10,10) << endl;
```

C:\Users\hira\Desktop\hello.exe

```
exp(1) = 2.71828
log(1) = 0
log10(100) = 2
isgreater(10,5) = 1
isless(10,5) = 0
isgreaterequal = 1
```

# Questions

- Write a C++ program to calculate e<sup>3</sup> using exp() and display the result.
- Write a C++ program to calculate the natural logarithm of 50 using log() and display the result.

23. Write a C++ program to calculate  $e^{\log(25)}$  using `exp()` and `log()`.
24. Write a C++ program to check whether 45 is greater than 30 using `isgreater()`.
25. Write a C++ program to check whether 15 is less than 25 using `isless()`.
26. Write a C++ program to check whether 20 is less than or equal to 20 using `islessequal()`.
27. Write a C++ program to calculate `exp(2)` and check whether the result is greater than 5.
28. Write a C++ program to calculate `log(100)` and check whether the result is less than 5.
29. Write a C++ program to calculate `exp(3)` and `log(10)`, and check whether the first result is greater than the second result.
30. Write a C++ program to calculate `exp(log(50))` and check whether the result is less than or equal to 50.

**Answer :** 20.0855, 3.9120, 25,1 (true),1 (true),1 (true),1 (true) ,1 (true), 1 (true),1 (true)

### 3. <cstring> (character string) OR string.h

Used to work with text data (sequence of characters). It's much easier than C-style character arrays.

| Name                   | Function use                                                                                                                                    | Example                                                                                        |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <code>Strlen()</code>  | Find length of string                                                                                                                           | <code>Strlen(hira) = 4</code>                                                                  |
| <code>Strcpy()</code>  | Copy complete string<br><b>Syntax :</b> <code>strcpy(destination,source);</code>                                                                | <code>a=hira,b;</code><br><code>strcpy(b,a) = hira</code><br><code>b=hira;</code>              |
| <code>Strncpy()</code> | Copy limited characters<br><b>Syntax :</b> <code>strncpy(destination,source,number)</code>                                                      | <code>A=hira,b;</code><br><code>Strncpy(b,a,2) = hi</code>                                     |
| <code>Strcat()</code>  | Concatenate or join strings<br><b>Syntax :</b> <code>strcat(desti, source)</code>                                                               | <code>a= hira , b= kumar;</code><br><code>strcat(a,b)=hirakumar</code>                         |
| <code>Strncat()</code> | Concatenate limited characters<br><b>Syntax:</b> <code>strncat(desti,sour,numb)</code>                                                          | <code>a=hira,b=kumar;</code><br><code>strncat(a,b,3)=hirakum</code>                            |
| <code>Strcmp()</code>  | Compare two string<br><b>Syntax :</b> <code>strcmp(1<sup>st</sup> stri, 2<sup>nd</sup> stri);</code><br>If $1^{\text{st}} == 2^{\text{nd}} = 0$ | <code>a=hira, b= kumar;</code><br><code>strcmp(a,b)= -1</code><br><code>strcmp(b,a) = 1</code> |

**HIRA KUMAR**

|           |                                                                                        |                                                            |
|-----------|----------------------------------------------------------------------------------------|------------------------------------------------------------|
|           | $1^{st} < 2^{nd}$ = any -ve value<br>$1^{st} > 2^{nd}$ = any +ve value                 |                                                            |
| Strncmp() | Compare first n characters<br>Syntax : strncmp(1 <sup>st</sup> , 2 <sup>nd</sup> , 3); | a=hira, b= kumar;<br>strncmp(a,b,3) =                      |
| Strchr()  | Find first occurrence of character                                                     | Char z[]="bananas"<br>Char *p=strchr(z,'a');<br>p = ananas |
| Strrchr() | Find last occurrence of character                                                      | Char z[]="bananas"<br>Char *p=strrchr(z,'a');<br>p = as    |
| Strstr()  | Find substring and print                                                               |                                                            |
| Memcpy()  | Copy memory                                                                            |                                                            |
| Memset()  | Set memory                                                                             |                                                            |

```
#include<cstring>
using namespace std;
int main()
{
    char a[10] = "hira";
    cout << "Total character length in " << a << " is : " << strlen(a) << endl;

    char b[10];
    strcpy(b,a);
    cout << "Copy other variable : " << b << endl;

    char c[10];
    strncpy(c,b,2);
    cout << "Copy n(number) character in other variable : " << c << endl;

    char e[10]="Kumar";
    strcat(a,e);
    cout << "join character is : " << a << endl;

    strncat(a,e,3);
    cout<< " join limited char : " << a;

    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
Total character length in hira is : 4
Copy other variable : hira
Copy n(number) character in other variable : hi
join character is : hiraKumar
join limited char : hiraKumarKum
```

```
#include<cstring>
using namespace std;
int main()
{
    char a[10] = "hira";
    cout << "Total character length in " << a << " is : " << strlen(a) << endl;

    char b[10];
    strcpy(b,a);
    cout << "Copy other variable : " << b << endl;
    cout << "a & b string are equal ? : " << strcmp(a,b) << endl;

    char c[10]="kumar";
    cout << "a & c are equal ? " << strcmp(c,a) << endl;
    cout << "a & c n number : " << strncmp(a,c,3) << endl;

    char z[10] = "bananas";
    char *p = strchr(z, 'a');
    cout << " p = " << p << endl;
    char *q = strrchr(z,'a');
    cout << " q = " << q << endl;

    char ab[20]="my name is hira";
    char *pp = strstr(ab,"name");
    cout << " pp = " << pp << endl;
}
```

Select C:\Users\hira\Desktop\hello.exe

```
Total character length in hira is : 4
Copy other variable : hira
a & b string are equal ? : 0
a & c are equal ? 1
a & c n number : -1
p = ananas
q = as
pp = name is hira
```

```
#include<iostream>
#include<conio.h>
#include<cstring>
using namespace std;
int main()
{
    char a[10] = "hira";
```

```

cout << " Total character length in " << a <<" is : " << strlen(a) << endl;

char b[10];
strcpy(b,a);
cout << "Copy other variable : " << b << endl;
cout << "a & b string are equal ? : " << strcmp(a,b) << endl;

char c[10]="kumar";
cout << "a & c are equal ? " << strcmp(c,a) << endl;
cout << " a & c n number : " << strncmp(a,c,3) << endl;

char z[10] = "bananas";
char *p = strchr(z, 'a');
cout << " p = " << p << endl;
char *q = strrchr(z,'a');
cout << " q = " << q << endl;

char ab[20]="my name is hira";
char *pp = strstr(ab,"name");
cout << " pp = " << pp << endl;

char cc[10];
strncpy(cc,b,2);
cout << "Copy n(number) character in other variable : " << c << endl;

char e[10]="Kumar";
strcat(a,e);
cout << "join character is : " << a << endl;

strncat(a,e,3);
cout<< " join limited char : " << a;

getch();

}

```

## 4. <ctime>

The <ctime> header file is used to work with date and time in c++.

| Function | Use                                                                                           |
|----------|-----------------------------------------------------------------------------------------------|
| time()   | Returns the current calendar time<br>Syntax : datatype variable;<br>Variable = time(nullptr); |

```
#include<iostream>
#include<conio.h>
#include<ctime>
using namespace std;
int main()
{
    time_t hira;
    hira = time(nullptr);
    cout << " Time is (in sec) : " << hira;
```

|         |                                                                             |
|---------|-----------------------------------------------------------------------------|
| ctime() | ctime() converts a time_t value into a human readable data and time string. |
|---------|-----------------------------------------------------------------------------|

```
#include<iostream>
#include<conio.h>
#include<ctime>
using namespace std;
int main()
{
    time_t hira;
    hira = time(nullptr);
    cout << ctime(&hira);
```

**Note:** Use time(nullptr). The output is: Time in seconds: 1789759352. This time is continuously calculated in seconds from **1 January 1970** because **1 January 1970, 00:00:00 UTC** was chosen as the standard starting point, called the **Unix Epoch**.

For example

1 Jan 1970 → 0 seconds

1 Jan 1970 + 1 second → 1

1 Jan 1970 + 60 seconds → 60

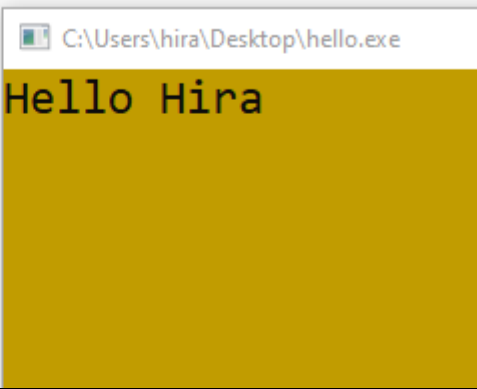
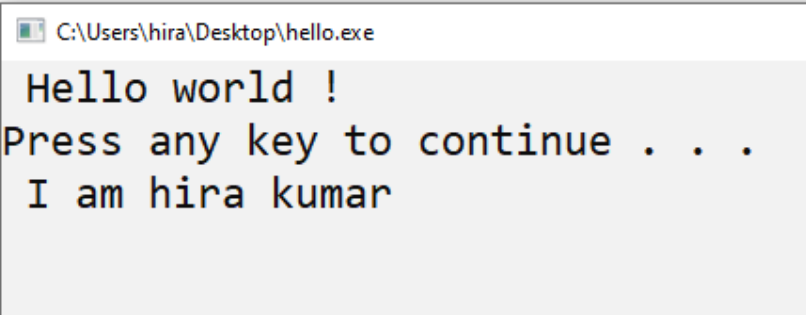
...

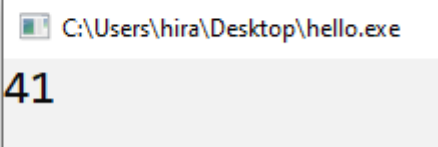
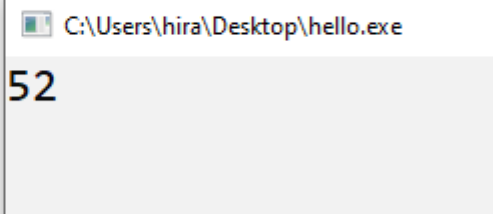
2026 → about 1.79 billion seconds

## 5. <cstdlib>

The <cstdlib> header provides general-purpose utility functions for c++ programs. It contains functions related to random numbers, string to number conversion, memory allocation, program termination, searching/sorting, text and background color change and mathematical utilities.

| Function | Use                                                                                     | Example                                                  |
|----------|-----------------------------------------------------------------------------------------|----------------------------------------------------------|
| system() | Change text/background color<br>Syntax :<br>system("color background_color text_color") | system("color 70");<br><br>or<br><br>system("color A4"); |

|  |                                                                                                                                                                                                                                                                                                                  |                                                                                      |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  | Common Windows color codes<br>0 : Black<br>1 : Blue<br>2 : Green<br>3 : Cyan<br>4 : Red<br>5 : Magenta<br>6 : Yellow/Brown<br>7 : White<br>8 : Gray<br>9 : Light Blue<br>10(A) : Light Green<br>11(B) : Light Cyan<br>12(C) : Light Red<br>13(D) : Light Magenta<br>14(E) : Light Yellow<br>15(F) : Bright White |                                                                                      |
|  | <pre>#include&lt;iostream&gt; #include&lt;conio.h&gt; #include&lt;cstdlib&gt; using namespace std; int main() {     system("color 60");      cout &lt;&lt; "Hello Hira"; }</pre>                                                                                                                                 |  |
|  | Clear console screen                                                                                                                                                                                                                                                                                             | system("cls");                                                                       |
|  | Pause and wait for a key                                                                                                                                                                                                                                                                                         | system("pause");                                                                     |
|  | <pre>#include&lt;iostream&gt; #include&lt;conio.h&gt; #include&lt;cstdlib&gt; using namespace std; int main() {     cout &lt;&lt; " Hello world !" &lt;&lt; endl;     system ("pause");     cout &lt;&lt; " I am hira kumar " &lt;&lt; endl; }</pre>                                                             |  |
|  | Display/change system date                                                                                                                                                                                                                                                                                       | system("date")                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                    |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Display/change system time                                                                         | system("date")  |
| <pre>#include&lt;iostream&gt; #include&lt;conio.h&gt; #include&lt;cstdlib&gt; using namespace std; int main() {     cout &lt;&lt; " Hello world !" &lt;&lt; endl;     system ("date");     cout &lt;&lt; " I am hira kumar " &lt;&lt; endl;     system("time"); }</pre>                                                                                                                                                        |                                                                                                    |                 |
| rand()                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Generate random number                                                                             | cout << rand(); |
| <pre>#include&lt;iostream&gt; #include&lt;conio.h&gt; #include&lt;cstdlib&gt; using namespace std; int main() {     cout &lt;&lt; rand(); }</pre>  <p>If you want generate number from b/w 1 to 10 then use formula :<br/>rand()%10+1<br/>If you want generate number from b/w 1 to 10 then use formula :<br/>rand()%100+1<br/>you want your given range :<br/>rand()%(y-x+1)+x (here x = starting value , y = last value)</p> |                                                                                                    |                 |
| srand()                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Initialize the random number generator use time().So the random sequence changes b/w program runs. |                 |
| <pre>#include&lt;iostream&gt; #include&lt;conio.h&gt; #include&lt;cstdlib&gt; #include&lt;ctime&gt; using namespace std; int main() {     srand(time(0));     cout &lt;&lt; rand()%100+1; }</pre>                                                                                                                                                                                                                            |                                                                                                    |                 |
| div()                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | div() performs integer division and returns both : quotient and remainder.                         |                 |

```
#include<iostream>
#include<conio.h>

using namespace std;
int main()
{
    div_t hira =div(22,4);
    cout << " quotient : " << hira.quot << endl;
    cout << " remainder : " << hira.rem <<endl;

    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
quotient : 5
remainder : 2
```

## 6. <string>

<string> is a standard C++ header file that provides the std::string class and operations for storing and manipulating strings.

```
#include<iostream>
#include<conio.h>
#include<string>
using namespace std;
int main()
{
    string n="hira";
    cout << " My name is " << n ;
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
My name is hira
```

Here:

- #include <string> → includes the C++ string library.
- string → represents a sequence of characters/text.
- n → a string variable.
- "hira" → string value.

| Function                     | Use                            | Example                                                   |
|------------------------------|--------------------------------|-----------------------------------------------------------|
| <b>n.length() / n.size()</b> | To find string length          | n="hira";<br>cout<<n.size(); // 4                         |
| <b>n.empty()</b>             | To check string blank or not   | n="";<br>if(n.empty()); // 1                              |
| <b>append()</b>              | Add two string                 | n="hira";<br>a.append("kumar");<br>cout<<a; // hira kumar |
| <b>+</b>                     | Add two string                 | string n="hira", n1="kr";<br>cout<< a+b; // hirakr        |
| <b>push_back()</b>           | Adds one character at the end. | string n="hir"<br>n.push_back('a');<br>cout<<n; // hira   |

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|                   |                                           |                                                            |
|-------------------|-------------------------------------------|------------------------------------------------------------|
| <b>pop_back()</b> | Remove last character                     | string n="hirag";<br>n.pop_back();<br>cout<< n; // hira    |
| <b>substr()</b>   | A part of the string coming<br>loose/find | string n="hirahiolal";<br>cout<<n.substr(4,7);//hiro       |
| <b>find()</b>     | Find position of<br>Character/string      | cout<< n.find("hiro"); // 4                                |
| <b>replace()</b>  |                                           | n.replace(7,3,"maharaj");<br>// hirahiromaharaj            |
| <b>erase()</b>    |                                           | n.erase(4,7); //<br>hiramaharaj                            |
| <b>insert()</b>   |                                           | n.insert(4," g "); //<br>hira g maharaj                    |
| <b>stoi()</b>     | String to integer                         | String a="123";<br>Int n= stoi(a);<br>Cout << n+10; // 133 |

```

#include<string>
using namespace std;
int main()
{
    string n="hira";
    cout << " My name is " << n << endl;
    cout << "My name is hira" << endl;
    cout << n.length() << endl;
    cout << "-----" << endl;
    if(n.empty())
        cout << " yes (1)" << endl;
    else
        cout<< "no (0)"<< endl;
    cout << "*****" << endl;
    n.append(" hiro ");
    cout << n << endl;
    cout << "-----" << endl;
    string nl="lal";
    cout << n+nl << endl;
    cout << "-----" << endl;
    n.push_back('a');
    cout << n << endl;
    cout << "*****" << endl;
    n.pop_back();
    cout << n << endl;
    cout << "-----" << endl;
    cout << n.substr(4,7) << endl;
    cout << "-----" << endl;
    cout<< n.find("hiro") << endl;

    cout << "*****" << endl;
    n.append("lal");
    cout << n << endl;
    cout << "*****" << endl;
    n.replace(10,3,"maharaj"); //same
    cout << n << endl;
    cout << "*****" << endl;
    n.erase(4,5); // same
    cout << n << endl;
    cout << "*****" << endl;
    n.insert(12," G ");
    cout << n << endl;
    cout << "*****" << endl;
    string a="123";
    int b=stoi(a);
    cout << b+10;
}

```

My name is hira

~~~~~

4

-----

no (0)

\*\*\*\*\*

hira hiro

hira hiro lal

-----

hira hiro a

\*\*\*\*\*

hira hiro

hiro

-----

5

~~~~~

\*\*\*\*\*

hira hiro lal

\*\*\*\*\*

hira hiro maharaj

\*\*\*\*\*

hira maharaj

\*\*\*\*\*

hira maharaj G

\*\*\*\*\*

133

## 7. <algorithm>

| Function         | Use                            | Example                                                                                |
|------------------|--------------------------------|----------------------------------------------------------------------------------------|
| <b>max()</b>     | Find max value                 | A=10, b=25;<br>Cout << max(a,b); // 25                                                 |
| <b>min()</b>     | Find min value                 | A=10, b=25;<br>Cout<< min(a,b); //10                                                   |
| <b>swap()</b>    | Swap two number                | A=10,b=25;<br>Swap(a,b);<br>Cout << a << b; // 25 10                                   |
| <b>sort()</b>    | Sort ascending order of array  | Int a[]={10,30,80,20};<br>sort(a,a+4);<br>for(int x : a)<br>cout << x ; // 10 20 30 80 |
| <b>reverse()</b> | Sort decsending order of array | Int a[]={10,30,80,20};                                                                 |

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|                |                           |                                                               |
|----------------|---------------------------|---------------------------------------------------------------|
|                |                           | reverse(a,a+4);<br>for(int x:a)<br>cout << x ; // 80 30 20 10 |
| <b>find()</b>  | Find number               | If(find(a,a+4,30)!= a+4)<br>Cout << "found";                  |
| <b>count()</b> | Frequency/repeat of value | Cout<< count(a,a+4,10); //1                                   |

```
#include<algorithm>
using namespace std;
int main()
{
int a=10, b=25;
cout << "Max value is " << max(a,b) << endl;
cout << "Min value is " << min(a,b) << endl;
swap(a,b);
cout << "swap value is a = " << a << ", b = " << b << endl;
int hira[]={10,30,80,20};
sort(hira,hira+4);
for(int z : hira)
cout << z << " " << endl;
reverse(hira,hira+4);
for(int z1 : hira)
cout << z1 << " ";
cout << count (hira,hira+4,10) << endl;
if(find(hira,hira+4,30)!=hira+4)
cout<<"found";
getch();
}
```

```
Max value is 25
Min value is 10
swap value is a = 25, b = 10
10
20
30
80
80 30 20 10 1
found
```

8. **<iomanip>**

9. **<cstdio>** → use printf(),scanf() in c++ language use this header file

10. **<vector>**

11. **<fstream>**

12. **<cctype>**

13. ....so many